

Experiment 4

CHARACTERISTICS OF A SCHOTTKY JUNCTION

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Objective:

- To study the D.C. characteristics of a Schottky barrier junction.
- To study the rectification and detection characteristics of a Schottky barrier diode.

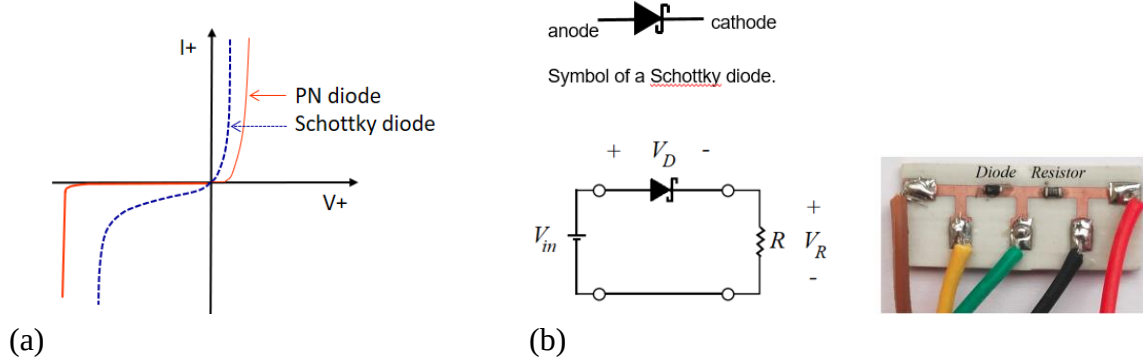


Fig. 1 (a) D.C. characteristics of a Schottky diode, and (b) the circuit used to obtain the D.C. characteristics.

Theory:

A Schottky diode is a specialized type of diode that employs the Schottky barrier formed at the junction between a metal and a semiconductor. This particular junction lacks a space-charge region, resulting in the absence of junction capacitance. Nevertheless, a small parasitic capacitance is present due to the diode's packaging. Additionally, it exhibits a low forward voltage. Schottky diodes find widespread use in high-speed switches, microwave power detectors, mixers, and multipliers operating at microwave to terahertz frequencies. These diodes take advantage of their non-linear characteristics for rectification and detection purposes. Refer to Fig. 1(a) for a depiction of a typical Schottky diode response. Fig. 1(b) shows the circuit used to attain its D.C. characteristics. The current in reverse bias conditions is several orders of magnitude higher than that of conventional *pn*-junction diodes. Thus, it is considered inferior as an rectifier. However, the forward knee voltage is significantly lower, rendering Schottky diodes suitable for detecting low-intensity microwave signals. The principle of power detection relies on rectifying the signal (AC component) through a diode and generating voltage as DC component. So, a bias circuit may not be necessary. Then, it is called a zero bias detector.

Fig. 2(a) shows a zero bias AM envelope detector circuit. The same circuit can be used as a rectifier. The RC circuit is used as a low pass filter. The diode current is given by

$$I = I_s \left[\exp\left(\frac{qV_D}{nkT}\right) - 1 \right].$$

Using Taylor series,

$$I = I_0 + \frac{\partial I}{\partial V} \partial V + \frac{1}{2} \frac{\partial^2 I}{\partial V^2} (\partial V)^2 + \dots$$

If the RF signal voltage is $\partial v = V_{RF} \cos(\omega_{RF} t)$, then the RF signal current is

$$i = I_{Bias} + \frac{\partial I}{\partial V} \bigg|_{bias} V_{RF} \cos(\omega_{RF} t) + \frac{1}{2} \frac{\partial^2 I}{\partial V^2} \bigg|_{bias} (V_{RF} \cos(\omega_{RF} t))^2 + \dots$$

For a zero-bias Schottky diode, the first term will be absent. Therefore, for a continuous wave microwave signal, the output DC follows the square term. However, when the input microwave signal is strong, the output may exhibit a linear variation with the input power level due to contributions from the higher-order terms.

$$\text{The diode conductance is } G_{RF} = \frac{\partial I}{\partial V} \bigg|_{bias} \approx \frac{q(I_{bias} + I_S)}{nkT}.$$

Now for amplitude modulated wave,

Considering input signal as $\partial V = V_{RF} (1 + m \sin(\omega_m t)) \cos(\omega_{RF} t)$, ω_m being the modulation frequency,

$$i = I_{Bias} + G_{RF} V_{RF} (1 + m \sin(\omega_m t)) \cos(\omega_{RF} t) + \frac{1}{2} \frac{\partial^2 I}{\partial V^2} \bigg|_{V_{bias}} V_{RF}^2 (1 + m \sin(\omega_m t))^2 \cos^2(\omega_{RF} t) \dots$$

After lowpass filtering, the modulated current is

$$I_m = \frac{1}{2} \frac{\partial^2 I}{\partial V^2} \bigg|_{V_{bias}} V_{RF}^2 \left(m \sin(\omega_m t) - \frac{m^2}{4} \cos(2\omega_m t) \right) \approx \frac{1}{2} \frac{\partial^2 I}{\partial V^2} \bigg|_{V_{bias}} m V_{RF}^2 \sin(\omega_m t).$$

The modulated signal depends on $m \sin(\omega_m t)$.

The conversion efficiency is related to the 2nd derivative of the I/V curve, i.e., the change in slope is

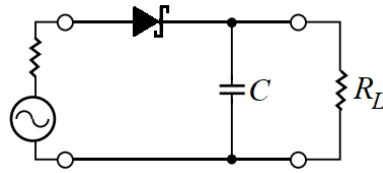
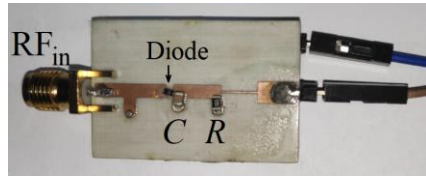
$$\frac{1}{2} \frac{\partial^2 I}{\partial V^2} \bigg|_{V_{bias}} = \frac{q^2 (I_{Bias} + I_S)}{2(nkT)^2} \approx \frac{q^2 I_{Bias}}{2(nkT)^2}.$$

The magnitude of the demodulated current is

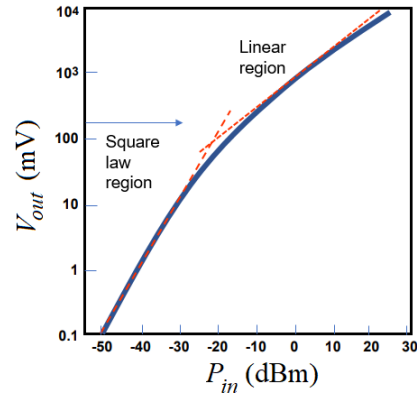
$$I_m = m V_{RF}^2 \frac{q^2 I_{Bias}}{2(nkT)^2} = \frac{q m V_{RF}^2}{2nkT} G_{RF} = \frac{q m P_{RF}}{nkT}. \quad \text{since, } P_{RF} = V_{RF}^2 \frac{G_{RF}}{2}.$$

Example: for square wave modulation with $m = 1$, $\frac{I_m}{P_{RF}} = \frac{qm}{nkT} = 38.6 \mu A / \mu W$ at 300K.

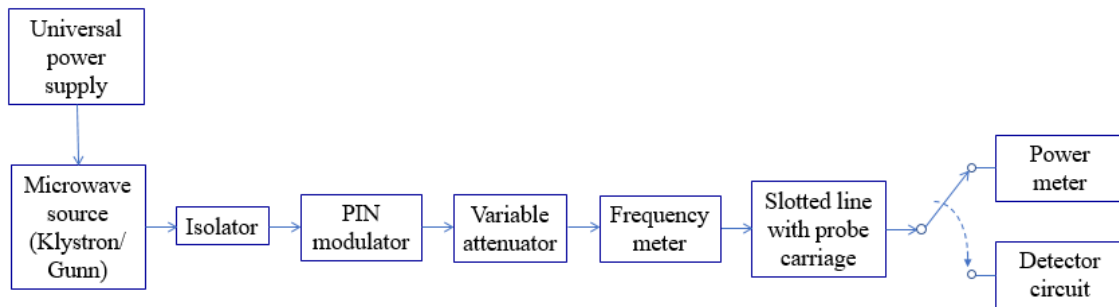
Fig. 2(a) shows a zero bias detector circuit. It does not require any D.C. power supply for its operation. When a continuous wave microwave signal is applied to its input, a D.C. voltage is generated at the output. The voltage varies with the input power. Also, the baseband signal is directly recovered for an amplitude modulated signal. Fig. 2(b) shows response of a rectifier/detector circuit. The detector behaves as square law device i.e. output voltage proportional to the square of input voltage for small input power. A key parameter for a diode detector is the slope of the transfer curve, which is called sensitivity of the detector γ , generally expressed in mV/ μ W. In the square law region, γ is usually constant when plotted against P_{in} (dBm). The sensitivity γ can be controlled by external bias current. It is also a function of temperature.



(a)



(b)



(c)

Fig. 2 (a) A simplified detector circuit, (b) output voltage of the detector for a continuous-wave input signal, (c) measurement set up to calibrate the detector.

Procedure:

D.C. characteristics:

1. Note down the maximum voltage and current rating of the diode.
2. Connect a variable D.C. voltage source as shown in Fig. 1(b).
3. Obtain both the forward and reverse bias characteristics as shown in Fig. 1(a).

Detector calibration:

1. Set up the measurement set up as shown in Fig. 2(c).
2. Connect the power meter and adjust the source frequency to the specified frequency of the detector circuit.
3. Note down the power reading shown by the power meter. Now replace the power meter by the detector circuit.
4. Vary the input power level by the calibrated attenuator and repeat the above step for 10-12 readings over -30 dBm to +10 dBm input power.
5. Obtain a plot as shown in Fig. 2(b) (Please note that it is a log-log plot). Identify the linear and square law regions.
6. Calculate and plot the diode sensitivity vs. P_{in} (dBm).

Rectifier mode: Here, the PIN modulation is not required. Repeat the above steps and plot $20\log_{10}(V_{out}(DC))$ vs P_{in} (dBm).